UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 18/02/2019*

EXERCISE

We observe N samples of the following discrete-time real random process:

$$X[n] = A + W[n], \quad n = 0, 1, \cdots, N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a random parameter, in particular it is modelled as a Gaussian random variable with known mean η_A and variance σ_A^2, $W[n]$ is additive white Gaussian noise (AWGN), independent of A, with known Power Spectral Density (PSD) $S_W\left(e^{j2\pi f}\right) = \sigma_W^2$.

(1) Derive the Autocorrelation Function (ACF) and the Power Spectral Density (PSD) of the w.s.s. process $X[n]$. Then, derive the joint probability density function (pdf) of the observed data vector $\mathbf{X} \triangleq [X[0] \quad X[1] \quad \cdots \quad X[N-1]]^T$ and of the random parameter A (this pdf can be expressed either in scalar form or in vector form, provide both). Finally, derive the marginal pdf of the observed data vector $\mathbf{X}$.

(2) Under the assumption that $\eta_A = 0$, derive the Maximum A-Posteriori (MAP) estimator of the amplitude A, its bias and Mean Square Error (MSE), and the pdf of the estimation error $\varepsilon \triangleq A - \hat{A}_{MAP}$. Express these quantities in terms of the parameter $\gamma \triangleq \sigma_A^2 / \sigma_W^2$ and establish whether the estimator is biased and consistent.

(3) Derive the Bayesian Cramér-Rao lower bound (BCRB) on the estimate of A and establish whether the MAP estimator is efficient or not.

Hints:

Woodbury's identity: $(\mathbf{I} + \gamma \mathbf{1}\mathbf{1}^T)^{-1} = \mathbf{I} - \dfrac{\gamma}{1 + \gamma \mathbf{1}^T \mathbf{1}} \mathbf{1}\mathbf{1}^T$

Determinant: $\left| \mathbf{I} + \gamma \mathbf{1}\mathbf{1}^T \right| = 1 + N\gamma$

Solution

(1) ACF:

$$R_X[m] = E\{X[n]X[n+m]\} = E\{(A+W[n])(A+W[n+m])\}$$
$$= E\{A^2\} + E\{W[n]W[n+m]\} = \eta_A^2 + \sigma_A^2 + \sigma_W^2\delta[m]$$

PSD:

$$S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\} = FT\{\eta_A^2 + \sigma_A^2 + \sigma_W^2\delta[m]\}$$
$$= 2\pi\left(\eta_A^2 + \sigma_A^2\right)\delta\left(e^{j2\pi f} - 1\right) + \sigma_W^2$$

PDF: Let's define $X_n \triangleq X[n]$. The pdf of the observed data vector conditioned to A is given by:

$$X_n | A \in N\left(A, \sigma_W^2\right), \quad \{X_n | A\}_{n=0}^{N-1} \text{ conditionally independent} \quad \Rightarrow \quad \mathbf{X} | A \in N\left(A\mathbf{1}, \sigma_W^2\mathbf{I}\right)$$

$$A \in N\left(\eta_A, \sigma_A^2\right)$$

The pdf in scalar format is:

$$f_{X,A}(\mathbf{x}, a) = f_{X|A}(\mathbf{x}|a) f_A(a) = \prod_{n=0}^{N-1} f_{X|A}(x_n|a) f_A(a)$$
$$= \left(\frac{1}{\sqrt{2\pi\sigma_W^2}}\right)^N \exp\left(-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x_n - a\right)^2\right) \cdot \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left(-\frac{(a-\eta_A)^2}{2\sigma_A^2}\right)$$

In vector format:

$$f_{X,A}(\mathbf{x}, a) = f_{X|A}(\mathbf{x}|a) f_A(a) = \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}} \exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x} - a\mathbf{1})^T(\mathbf{x} - a\mathbf{1})\right) \cdot \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left(-\frac{(a-\eta_A)^2}{2\sigma_A^2}\right)$$

Pdf of **X** in vector format: $\mathbf{X} = A\mathbf{1} + \mathbf{W}$, **X** is a Gaussian vector, since A and **W** are jointly Gaussian distributed: $\mathbf{X} \in N\left(\boldsymbol{\eta}_X, \mathbf{C}_X\right)$.

$$\boldsymbol{\eta}_X = E\{\mathbf{X}\} = E\{A\mathbf{1} + \mathbf{W}\} = E\{A\}\mathbf{1} + E\{\mathbf{W}\} = \eta_A\mathbf{1}$$

$$\mathbf{C}_X = E\left\{(\mathbf{X}-\eta_A\mathbf{1})(\mathbf{X}-\eta_A\mathbf{1})^T\right\} = E\left\{(A-\eta_A)^2\right\}\mathbf{1}\mathbf{1}^T + E\left\{\mathbf{W}\mathbf{W}^T\right\} = \sigma_A^2\mathbf{1}\mathbf{1}^T + \sigma_W^2\mathbf{I}$$

$$f_X(\mathbf{x}) = \frac{1}{(2\pi)^{N/2}\sqrt{\left|\sigma_A^2\mathbf{1}\mathbf{1}^T+\sigma_W^2\mathbf{I}\right|}}\exp\left(-\frac{1}{2}(\mathbf{x}-\eta_A\mathbf{1})^T\left(\sigma_A^2\mathbf{1}\mathbf{1}^T+\sigma_W^2\mathbf{I}\right)^{-1}(\mathbf{x}-\eta_A\mathbf{1})\right)$$

$$= \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}\sqrt{\left|\gamma\mathbf{1}\mathbf{1}^T+\mathbf{I}\right|}}\exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x}-\eta_A\mathbf{1})^T\left(\gamma\mathbf{1}\mathbf{1}^T+\mathbf{I}\right)^{-1}(\mathbf{x}-\eta_A\mathbf{1})\right)$$

where $\gamma \triangleq \sigma_A^2/\sigma_W^2$.

The inverse of the covariance matrix can be derived by using the *Woodbury's identity*:

$$(\mathbf{I}+\gamma\mathbf{1}\mathbf{1}^T)^{-1} = \mathbf{I} - \frac{\gamma}{1+\gamma\mathbf{1}^T\mathbf{1}}\mathbf{1}\mathbf{1}^T = \mathbf{I} - \frac{\gamma}{1+N\gamma}\mathbf{1}\mathbf{1}^T$$

The determinant is given by: $\left|\gamma\mathbf{1}\mathbf{1}^T+\mathbf{I}\right| = 1+N\gamma$. Hence, the pdf of $\mathbf{X}$ is given by:

$$f_X(\mathbf{x}) = \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}\sqrt{1+N\gamma}}\exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x}-\eta_A\mathbf{1})^T\left(\mathbf{I}-\frac{\gamma}{1+N\gamma}\mathbf{1}\mathbf{1}^T\right)(\mathbf{x}-\eta_A\mathbf{1})\right)$$

(2) MAP estimator A for $\eta_A = 0$.

$$f_{X,A}(\mathbf{x},a) = f_{X|A}(\mathbf{x}|a)f_A(a) = \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}}\exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x}-a\mathbf{1})^T(\mathbf{x}-a\mathbf{1})\right)\cdot\frac{1}{\sqrt{2\pi\sigma_A^2}}\exp\left(-\frac{a^2}{2\sigma_A^2}\right)$$

$$\hat{A}_{MAP} = \arg\max_a\left(\ln f_{X,A}(\mathbf{x},a)\right) = \arg\max_a\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right)$$

$$\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a) = \kappa - \frac{1}{2\sigma_W^2}(\mathbf{x}-a\mathbf{1})^T(\mathbf{x}-a\mathbf{1}) - \frac{a^2}{2\sigma_A^2}$$

$$= \kappa - \frac{1}{2\sigma_W^2}\mathbf{x}^T\mathbf{x} - \frac{Na^2}{2\sigma_W^2} + \frac{1}{2\sigma_W^2}a\mathbf{1}^T\mathbf{x} + \frac{1}{2\sigma_W^2}a\mathbf{x}^T\mathbf{1} - \frac{a^2}{2\sigma_A^2}$$

$$= \kappa - \frac{1}{2\sigma_W^2}\mathbf{x}^T\mathbf{x} + \frac{1}{\sigma_W^2}\left(\mathbf{1}^T\mathbf{x}\right)a - \left(\frac{N}{2\sigma_W^2}+\frac{1}{2\sigma_A^2}\right)a^2$$

$$\frac{d}{da}\left(\ln f_{X|A}(\mathbf{x}|a)+\ln f_A(a)\right)=\frac{N}{\sigma_W^2}\left(\frac{1}{N}\mathbf{1}^T\mathbf{x}\right)-\left(\frac{N}{\sigma_W^2}+\frac{1}{\sigma_A^2}\right)a=0$$

$$\hat{A}_{MAP}=\left(\frac{N}{\sigma_W^2}+\frac{1}{\sigma_A^2}\right)^{-1}\frac{N}{\sigma_W^2}\left(\frac{1}{N}\mathbf{1}^T\mathbf{x}\right)=\frac{\sigma_A^2 N}{\sigma_A^2 N+\sigma_W^2}\left(\frac{1}{N}\mathbf{1}^T\mathbf{x}\right)=\frac{\gamma N}{\gamma N+1}\left(\frac{1}{N}\mathbf{1}^T\mathbf{x}\right)$$

$$\hat{A}_{MAP}=\alpha\left(\frac{1}{N}\mathbf{1}^T\mathbf{x}\right)=\frac{\alpha}{N}\sum_{n=0}^{N-1}X[n],\quad \text{where}\ \ \alpha\triangleq\frac{N\gamma}{1+N\gamma}$$

Hence, the estimator is **linear**. Since the estimate is a linear function of the Gaussian distributed data vector **X**, the estimate is Gaussian distributed, and consequently the estimation error is Gaussian distributed, with mean and variance given by:

$$E\left\{\hat{A}_{MAP}\right\}=E\left\{\frac{\alpha}{N}\mathbf{1}^T\mathbf{x}\right\}=\frac{\alpha}{N}\mathbf{1}^T E\{\mathbf{x}\}=\frac{\alpha}{N}\mathbf{1}^T\boldsymbol{\eta}_X=0\quad\Rightarrow\quad E\left\{A-\hat{A}_{MAP}\right\}=E\{A\}-E\left\{\hat{A}_{MAP}\right\}=0$$

Hence, the estimator is **unbiased**.

$$\begin{aligned}MSE\left\{\hat{A}_{MAP}\right\}&=E\left\{\left(A-\hat{A}_{MAP}\right)^2\right\}=E\left\{\left(A-\frac{\alpha}{N}\mathbf{1}^T\mathbf{x}\right)^2\right\}=E\left\{\left(A-\frac{\alpha A}{N}\mathbf{1}^T\mathbf{1}+\frac{\alpha}{N}\mathbf{1}^T\mathbf{W}\right)^2\right\}\\&=E\left\{\left((1-\alpha)A+\frac{\alpha}{N}\mathbf{1}^T\mathbf{W}\right)^2\right\}=(1-\alpha)^2\sigma_A^2+\frac{\alpha^2}{N^2}\mathbf{1}^T E\left\{\mathbf{W}\mathbf{W}^T\right\}\mathbf{1}=(1-\alpha)^2\sigma_A^2+\frac{\alpha^2}{N}\sigma_W^2\\&=\frac{1}{(1+N\gamma)^2}\sigma_A^2+\frac{N\gamma^2}{(1+N\gamma)^2}\sigma_W^2=\frac{\sigma_W^2\gamma}{1+N\gamma}=\alpha\frac{\sigma_W^2}{N}\end{aligned}$$

$$\lim_{N\to\infty}MSE\left\{\hat{A}_{MAP}\right\}=\lim_{N\to\infty}\left(\frac{\sigma_W^2\gamma}{1+N\gamma}\right)=0$$

Hence, the estimator is **consistent**.

PDF of the estimation error: $\varepsilon=A-\hat{A}_{MAP}\in N\left(0,\alpha\frac{\sigma_W^2}{N}\right)$

(3) BCRB on the estimate of *A*:

$$\frac{d^2 \ln f_{X,A}(\mathbf{x},a)}{da^2} = \frac{d^2}{da^2}\left(\ln f_{X|A}(\mathbf{x}|a) + \ln f_A(a)\right) = \frac{d}{da}\left[\frac{N}{\sigma_W^2}\left(\frac{1}{N}\mathbf{1}^T\mathbf{x}\right) - \left(\frac{N}{\sigma_W^2} + \frac{1}{\sigma_A^2}\right)a\right] = -\left(\frac{N}{\sigma_W^2} + \frac{1}{\sigma_A^2}\right)$$

$$BCRB(A) = \frac{1}{-E\left\{\dfrac{d^2 \ln f_{X,A}(\mathbf{x},a)}{da^2}\right\}} = \left(\frac{N}{\sigma_W^2} + \frac{1}{\sigma_A^2}\right)^{-1} = \frac{\sigma_A^2\sigma_W^2}{\sigma_W^2 + N\sigma_A^2} = \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2}\cdot\frac{\sigma_W^2}{N} = \alpha\frac{\sigma_W^2}{N} = MSE\left\{\hat{A}_{MAP}\right\}$$

Hence, the estimator is **efficent**.